

Sentiment Analysis: Main Approaches

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Sentiment Analysis → recap

- ▶ Motivation
 - ▶ Why doing it?
 - ▶ For whom?
 - ▶ Examples of a full analysis

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- ▶ Examples of target of opinions

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- ▶ Factual vs. subjective data
- ▶ Bing Liu's model

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- ▶ Motivation
 - ▶ Why doing it?
 - ▶ For whom?
 - ▶ Examples of a full analysis
- ▶ Examples of target of opinions
- ▶ Factual vs. subjective data
- ▶ Bing Liu's model
- ▶ Granularity levels
 - ▶ Document-level
 - ▶ Sentence-level
 - ▶ Feature-level

Sentiment Analysis → recap

On 25th Oct. 2018 , John Doe wrote:

“ I bought the new iPhone a few days ago. It was such a nice phone . The touch screen was really cool . The voice quality was clear too. Although the battery life is not long , that is ok for me. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive , and wanted me to return it to the shop.”

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- ▶ f_{jk} : phone, touch screen, voice quality, battery life, phone

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- ▶ so_{ijkl} : positive, positive, positive, negative, negative

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- ▶ opinion holder h_i : I, I, I, I, mother

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- ▶ o_j : iPhone
- ▶ f_{jk} : phone, touch screen, voice quality, battery life, phone
- ▶ so_{ijkl} : **positive**, **positive**, **positive**, **negative**, **negative**
- ▶ opinion holder h_i : I, I, I, I, mother
- ▶ time t_I : 25/10/2018

Sentiment Analysis granularity → feature level 2/2

1. Identify **entities** or **objects**
 - ▶ Given a set Q of **seed entities** of class C , and a set D of **candidate entities**, determine which of the entities in D belong to C .
 - ▶ binary classification problem
2. Extract object **features** that have been commented on by the **opinion holder**
 - ▶ Frequency-based methods [Hu and Liu, 2004]
 - ▶ Challenge: infrequent feature extraction
3. Group similar **features**
 - ▶ e.g. screen and touch screen
 - ▶ e.g. power usage and battery life
4. Classify the opinions as **positive**, **neutral** or **negative**
 - ▶ detailed later
5. [optional] Produce a summary of all feature-based opinions

Sentiment Analysis: result

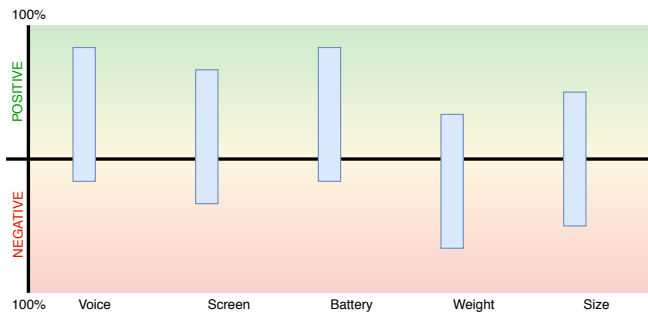
Component	Polarity	Comment
Feature1 <u>screen</u> touch screen	Positive	The touch screen was really cool. The touch screen was so easy to use and can do amazing things. The screen is so fluid!
	Negative	The screen is easily scratched. It's very difficult to remove finger marks from the touch screen.
Feature2 <u>battery life</u>	Positive	The battery stood for 2 days. The battery lasts very long.
	Negative	I had to charge it after 2 hours... ...

Learning objectives

- ▶ Identify possible applications of SA and related tasks
- ▶ Define the main challenges in SA
- ▶ Explain the lexicon-based approaches for SA

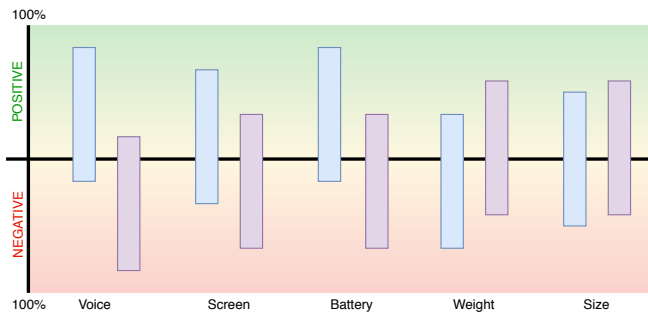
SA examples of possible analysis

Sentiment Analysis: opinion observer



Summary of reviews for smartphone 1
[Liu et al., 2005]

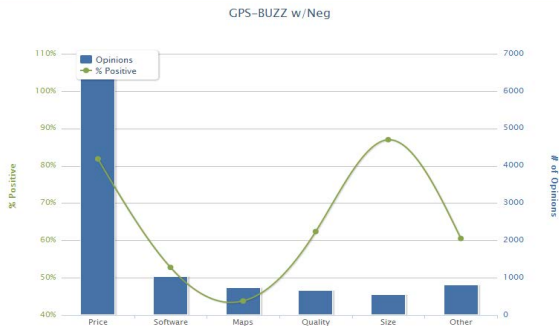
Sentiment Analysis: opinion observer



Comparison of summaries of reviews for two smartphones
[Liu et al., 2005]

Sentiment Analysis: indicators

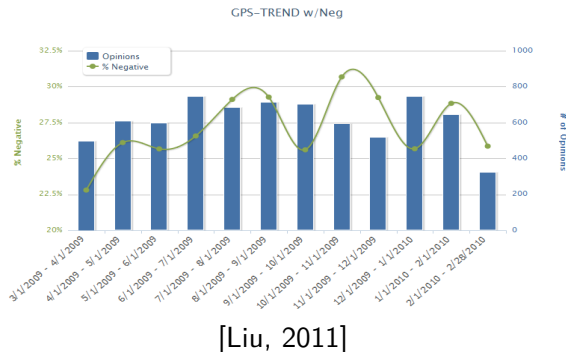
- ▶ absolute number of expressed opinions and proportion



[Liu, 2011]

Sentiment Analysis: indicators

► evolution of opinions across time



SA challenges

Sentiment Analysis: challenges

On 4th Nov. 2018 , John Doe wrote:

This past Saturday, I bought a Nokia phone and my girlfriend bought a Motorola phone with Bluetooth . We called each other when we got home. The voice on my phone was not so clear , worse than my previous phone. The battery life was short too.

My girlfriend was quite happy with her phone.

I wanted a phone with good sound quality . So my purchase was a real disappointment . I returned the phone yesterday.

Sentiment Analysis: challenges

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I returned the phone yesterday.

→ **Identify all the components of an opinion**

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- Identify all the components of an opinion
- Identify the relations

Sentiment Analysis: challenges

Discover all quintuples $(o_j, f_{jk}, so_{ijkl}, h_i, t_l)$ in a document

- ▶ o_j – the target **object** or **entity**:
 - ▶ **Named Entity Recognition.**
 - ▶ **Coreference resolution.**
- ▶ f_{jk} – a **feature** or **aspect**: **Information Extraction** (more about that later...).
- ▶ so_{ijkl} – **sentiment value**: **Sentiment Identification**
- ▶ h_i – **sentiment holder**: **Information/Meta-Data Extraction**
- ▶ t_l – **time**: **Information/Meta-Data Extraction**

In addition:

- ▶ **Relation Extraction**
- ▶ **Synonym match**: e.g. "voice" == "sound quality"

Sentiment Analysis: Lexicon-based approaches

Sentiment Analysis: two main approaches

- ▶ **Lexicon-based**

- ▶ Binary
- ▶ Gradable

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- ▶ **Corpus-based**

- ▶ Naive Bayes
- ▶ **Deep Learning**

Sentiment Analysis: lexicon-based

Principle

- ▶ Use a lexicon of opinion/emotion words → words with **polarity**
- e.g. good, bad, horrible, great, etc.

Rule-based sentiment classifier at **sentence** or **document** level

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Rule-based sentiment classifier at **sentence** or **document** level

1. Rule-based **subjectivity classifier**

- ▶ text is **subjective** if it has n words from the emotion lexicon (n is fixed by expert)
- ▶ **objective** otherwise

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- ▶ text is **subjective** if it has n words from the emotion lexicon (n is fixed by expert)
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2. Rule-based **sentiment classifier**

- ▶ applied on **subjective** text only
- ▶ **count** the number of positive and negative word/phrases in the text
- ▶ text is
 - ▶ **negative** if more negative than positive
 - ▶ **positive** if more positive than negative
 - ▶ **neutral** otherwise

Sentiment Analysis: lexicon-based → binary

Rule-based **sentiment classifier** at **feature** level

- ▶ Assume **features** can be identified in a previous step → battery, phone, screen
- ▶ Identify **emotion** associated with those **features**
- ▶ count **negative** and **positive** emotion words/phrases from the lexicon
- ▶ feature is
 - ▶ **negative** if more negative than positive
 - ▶ **positive** if more positive than negative
 - ▶ **neutral** otherwise

Sentiment Analysis: lexicon-based $\rightarrow$ binary

Rule-based **sentiment classifier** at **feature** level

- ▶ Simple approach:
 - ▶ **input**: a pair (f, S) where f is a product feature and S is a sentence containing f
 - ▶ **output**: a label in either **negative**, **positive** or **neutral**.

Sentiment Analysis: lexicon-based → binary

Rule-based **sentiment classifier** at **feature** level

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 - ▶ **input**: a pair (f, S) where f is a product feature and S is a sentence containing f
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- ▶ Protocol: consider $S = w_1, \dots, w_N$ the sentence containing f , with N its length
 1. select the **emotion** words w_i in S
 2. assign **orientations** to each of these words w_i
 - ▶ **negative** → -1
 - ▶ **positive** → +1
 - ▶ **neutral** → 0
 3. **sum up** the orientation and **assign** a label to (f, S) accordingly

Sentiment Analysis: lexicon-based → binary

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- ▶ more advanced strategies:
 - ▶ split the sentence in parts using **discourse connectives/markers**
 - ▶ focus on markers that may introduce a change in the sentiment
→ "but", "except that"

Sentiment Analysis: lexicon-based → binary (caveats)

Certain words have context-independent orientations, e.g. “great”

Other emotion words have **context-dependent orientations**, e.g.

- ▶ “**small power consumption**” is positive but **small capacity** is negative
- ▶ “**consume valuable resources**” is negative but **consume disgusting waste** is positive

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One has to deal with **negation**, e.g.:

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One has to deal with **intensifiers**:

- ▶ **very good** is more positive than **good**
- ▶ **extremely boring** is more negative than **boring** or **very boring**

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→ need a more **fine-grained sentiment information** in lexicon and add **additional rules**.

Sentiment Analysis: two main approaches

- ▶ **Lexicon-based**

- ▶ Binary
- ▶ **Gradable**

- ▶ **Corpus-based**

- ▶ Naive Bayes
- ▶ Deep Learning

Sentiment Analysis: lexicon-based → gradable

Principle

- ▶ Use of **ranges of sentiment** instead of a **binary system**
 - ▶ deal with **intensifiers** like: absolutely, utterly, completely, totally, nearly, virtually, essentially, mainly, almost, ...
- e.g.: **absolutely awful**

Sentiment Analysis: lexicon-based → gradable

Principle

- ▶ Use of **ranges of sentiment** instead of a **binary system**
- ▶ deal with **intensifiers** like: absolutely, utterly, completely, totally, nearly, virtually, essentially, mainly, almost, ...
- e.g.: **absolutely awful**

Grade adverbs like:

- ▶ Very, little, dreadfully, extremely, fairly, hugely, immensely, intensely, rather, reasonably, slightly, unusually, ...
- e.g.: **immensely beautiful**

Sentiment Analysis: lexicon-based → gradable

Rule-based **gradable sentiment classifier**

- ▶ **Classify** general **valence** of a text based on **the level of emotional content**
- ▶ Level of emotional content given by:

Sentiment Analysis: lexicon-based → gradable

Rule-based **gradable sentiment classifier**

- ▶ **Classify** general **valence** of a text based on **the level of emotional content**
- ▶ Level of emotional content given by:
 1. the **lexicon**: word list with pre-assigned emotional weights

bore	-3
boring	-3
bother	-1
brave	3
bright	2
brilliant	2
broke	-1
brutal	-3
burden	-1
calm	2
care	2
cared	2
carefree	2
careful	3
careless	-2
careless	2
caring	3
casual	2

- ▶ **negative dimension:** $C_{neg} \in -5, \dots, -1$
- ▶ **positive dimension:** $C_{pos} \in +1, \dots, +5$
- ▶ **neutral (0)** is ignored

Sentiment Analysis: lexicon-based → gradable

Rule-based **gradable sentiment classifier**

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2. additional **general rules**: word list with pre-assigned emotional weights

- ▶ **negation rule**

- ▶ **capitalization rule**

- ▶ **intensifier rule**

- ▶ **diminisher rule**

- ▶ **exclamation rule**

- ▶ **emoticon rule**

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caring	3
casual	2

Sentiment Analysis: lexicon-based → gradable

2. additional **general rules**: word list with pre-assigned emotional weights
- ▶ **negation rule**: **reduce** and **invert** the weight when the word "not" is in the neighbourhood
 - ▶ Ex.: "I am not good today" and in the lexicon:
 $\text{emotion}(\text{good}) = 3$
 - ▶ **reduce** the value by 1 and **inverse** the polarity
 - $\text{new emotion}(\text{good}) = (+3 - 1) * -1 = -2$
- ▶ **capitalization rule**: **strengthen** the weight when written in capitals
 - ▶ Ex.: "I am GOOD today"
 - $\text{new emotion}(\text{good}) = (+3 + 1) = +4$
 - ▶ Ex.: "I am AWFUL today" and in the lexicon:
 $\text{emotion}(\text{awful}) = -4$
 - $\text{new emotion}(\text{awful}) = (-4 - 1) = -5$

Sentiment Analysis: lexicon based: gradable

2. additional **general rules**: word list with pre-assigned emotional weights
 - ▶ **intensifier rule**: **strengthen** the weight of the modified word
 - ▶ Needs a list of intensifiers with their weights
 - similar to the word lexicon: $\text{weight}(\text{very}) = 1$, $\text{weight}(\text{extremely}) = 2$, etc.
 - ▶ the weight is **added** to **positive** terms
 - ▶ the weight is **subtracted** to **negative** terms
 - ▶ Ex.: "I am feeling very good"
 - **new emotion(good)** = $(+3 +1) = +4$
 - ▶ Ex.: "This was an extremely boring game" and in the lexicon:
emotion(boring) = -3
 - **new emotion(boring)** = $(-3 -2) = -5$

Sentiment Analysis: lexicon-based → gradable

2. additional **general rules**: word list with pre-assigned emotional weights

▶ **diminisher rule**: **weaken** the weight of the modified word

▶ Needs a list of diminishers with their weights

→ similar to the word lexicon: $\text{weight}(\text{somewhat}) = 1$, $\text{weight}(\text{slightly}) = 1$, etc.

▶ the weight is **subtracted** to **positive** terms

▶ the weight is **added** to **negative** terms

▶ Ex.: "I am somewhat good"

→ **new emotion(good)** = $(+3 -1) = +2$

▶ Ex.: "This was an slightly boring game" and in the lexicon:

emotion(boring) = -3

→ **new emotion(boring)** = $(-3 +1) = -2$

Sentiment Analysis: lexicon-based → gradable

- 2. additional **general rules**: word list with pre-assigned emotional weights
 - ▶ **exclamation rule**: **strengthen** the weight of the modified word (similar to **intensifiers**)
 - ▶ Needs a list of exclamations with their weights
 - similar to the word lexicon: $\text{weight}(!!!) = 2$, $\text{weight}(!!!!!!!!!!) = 3$, etc.
 - ▶ the weight is **added** to **positive** terms
 - ▶ the weight is **subtracted** to **negative** terms
 - ▶ Ex.: "Great show!!!" and in the lexicon: $\text{emotion}(\text{great}) = +3$
 - $\text{new emotion}(\text{great}) = (+3 + 2) = +5$

Sentiment Analysis: lexicon-based → gradable

2. additional **general rules**: word list with pre-assigned emotional weights
- ▶ **emoticon rule**: provide a weight to each emoticon (similar to words in **lexicon**)
 - ▶ Needs a list of emoticons with their weights
 - ▶ **emotion(😊) = +2**, **emotion(😞) = -2**
 - ▶ Ex1.: "I can't believe this product 😊" → **emotion(Ex1.) = +2**
 - ▶ Ex2.: "I can't believe this product 😞" → **emotion(Ex2.) = -2**

Sentiment Analysis: lexicon-based → gradable

The **valence** of the text is the sum of the weights of the emotional words
Consequently:

- ▶ if $|C_{pos}| < |C_{neg}|$ then **emotion(text) = negative**
- ▶ if $|C_{pos}| > |C_{neg}|$ then **emotion(text) = positive**
- ▶ if $|C_{pos}| = |C_{neg}|$ then **emotion(text) = neutral**

- ▶ Ex1.: text = "He is brilliant but boring", emotion(brilliant) = +2, emotion(boring) = -3
→ **valence(text) = +2 - 3 = -1 → negative**
- ▶ Ex2.: text = "I am not good today", emotion(good) = +2,
negation rule
→ **valence(text) = (+2 - 1) * -1 = -1 → negative**
- ▶ Home exercises:
 - ▶ Ex3. "I am not GOOD today"
 - ▶ Ex4. "I am so surprised by this product!!! ☹"

Sentiment Analysis: lexicon-based

► Advantages

- Works effectively with different texts: forums, blogs, etc.
- Language independent → only need a lexicon with emotion weights
- No training data required
- Easily extendible: simply add new entries in the dictionary

► Disadvantages

- The lexicon of emotion word is a resource made by experts
 - costly to create and maintain
- Needs frequent updates to incorporate new words/abbreviations
- Not robust to spelling errors
- A dataset from MySpace has 95% of comments containing at least one spelling error
- Static resource

Sentiment Analysis: lexicon-based

How to obtain lexica of emotion words?

1. Collect relevant words/phrases expressing sentiment
 2. Determine the emotion of these subjective texts
- Can be done **manually**
- ▶ word list with pre-assigned emotional weights
- Can be done **semi-automatically**
- ▶ require a dictionary of **seed emotion words**
 - ▶ **dictionary-based**
 - ▶ find synonyms/antonyms using linguistic resources like e.g. WordNet
 - <https://wordnet.princeton.edu> and <http://projects.illc.uva.nl/EuroWordNet>
 - <http://globalwordnet.org/resources/wordnets-in-the-world/>
 - ▶ **corpus-based**
 - ▶ find synonyms/antonyms in corpora

Sentiment Analysis: lexicon-based

What do lexica of emotion words contain?

▶ Adjectives

- ▶ **positive**: honest, important, mature, large, patient, ...
- ▶ **negative**: harmful, hypocritical, inefficient, insecure, ...

▶ Verbs

- ▶ **positive**: praise, love, ...
- ▶ **negative**: blame, criticize, ...

▶ Nouns

- ▶ **positive**: pleasure, enjoyment, ...
- ▶ **negative**: pain, criticism, ...

▶ Phrases: for **collocations**, also an alternative to **intensifiers**

- ▶ **positive**: "very efficient", "low cost", ...
- ▶ **negative**: "many problems", "lot of bugs", ...

Sentiment Analysis: lexicon based: manually created resources

Manually created resources by experts

- ▶ **SentiWordNet** database:
<http://ontotext.fbk.eu/sentiwn.html>
 - ▶ **Wordnet**: words grouped in sets of synonyms (**synsets**)
 - ▶ with semantic relations between them: synonyms, antonyms, hypernyms, etc.
 - sentiment score added **negative**, **positive** and **neutral**
- ▶ **Linguistic Inquiry and Word Count (LIWC) lexicon**
 - ▶ made by psychologists
 - ▶ words with several **emotional and other dimensions**
- ▶ **General Inquirer**
 - ▶ terms with various types of **negative** or **positive** semantic orientation
 - <http://www.wjh.harvard.edu/~inquirer/homecat.htm>

SentiWordNet



SentiWordNet

ADJECTIVE

P: 0.25 O: 0.5 N: 0.25

terrific1 01513619

very great or intense; "a terrific noise"; "a terrific thunderstorm storm"; "fought a terrific battle"

Feedback on SentiWordNet values: [They are OK.](#) [Suggest your values.](#)

P: 0.75 O: 0.25 N: 0

wondrous#1 wonderful#1 tremendous#2 **terrific#2** rattling#1 marvelous#1 marvelous#1 howling#1 grand#4 fantastic#2 01676517

extraordinarily good or great ; used especially as intensifiers; "a fantastic trip to the Orient"; "the film was fantastic!"; "a howling success"; "a marvelous collection of rare books"; "had a rattling conversation about politics"; "a tremendous achievement"

Feedback on SentiWordNet values: [They are OK.](#) [Suggest your values.](#)

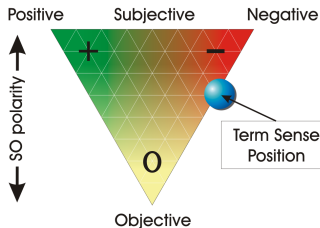
P: 0 O: 0.375 N: 0.625

terrifying#1 **terrific#3** 00196449

causing extreme terror; "a terrifying wall"

Feedback on SentiWordNet values: [They are OK.](#) [Suggest your values.](#)

← PN polarity →



Linguistic Inquiry and Word Count (LIWC) lexicon

Table 1. LIWC2007 Content Categories of Affective, Cognitive and Perceptual Processes, with Subcategories, Abbreviation and Examples (Pennebaker, Booth, & Francis, 2007).

Category	Abbreviation	Examples
Affective processes	affect	Happy, cried, abandon
Positive emotion	posemo	Love, nice, sweet
Negative emotion	negemo	Hurt, ugly, nasty
Anxiety	anx	Worried, fearful, nervous
Anger	anger	Hate, kill, annoyed
Sadness	sad	Crying, grief, sad
Cognitive processes	cognitive	Cause, know, ought
Insight	insight	Think, know, consider
Causation	cause	Because, effect, hence
Perceptual processes	percept	Observing, heard, feeling
See	see	View, saw, seen
Hear	hear	Listen, hearing
Feel	feel	Feels, touch

Brönnimann et al. 2013 [Brönnimann et al., 2013]

Sentiment Analysis: lexicon-based → semi-automatic methods

Semi-automatically created from **seed words**

- ▶ start with **seed negative** and **positive** words
 - ▶ **dictionary-based**: synonyms/antonyms in dictionaries like WordNet
 - ▶ **corpus-based**: search on large corpora, like the web, for **patterns**
 - ▶ "beautiful and" (+)
 - ▶ "low cost but" (-)
 - ▶ "very nice and" (+)

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Very nice and **perfect**

★★★★★ Note : 5 - Critique de a TripAdvisor user - Prix : \$ (Based on Average Nightly Rates for a Standard Room from our Partners)

Very nice and **comfortable**

★★★★★ Note : 5 - Critique de a TripAdvisor user - Prix : \$ (Based on Average Nightly Rates for a Standard Room from our Partners)

Very nice and **quiet apartment near the beach**

★★★★★ Note : 4,7 - 6 avis

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Very expensive but **excellent food and wine**

★★★★★ Note : 5 - Critique de a TripAdvisor user

Very expensive but **good food.**

★★★★★ Note : 3 - Critique de a TripAdvisor user

Very expensive but **disappointing..**

★★★★★ Note : 1 - Critique de a TripAdvisor user

Learning objectives

- ▶ Define the feature-level granularity level for SA
- ▶ Identify possible applications of SA and related tasks
- ▶ Define the main challenges in SA
- ▶ Explain the lexicon-based approaches for SA

Reading material

- ▶ Bing Liu (2012).
Sentiment Analysis and Opinion Mining.
Morgan and Claypool Publishers. Draft on line at:
<https://www.cs.uic.edu/~liub/FBS/SentimentAnalysis-and-OpinionMining.pdf>
- ▶ C. J. Hutto, Eric Gilbert (2014).
VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text
<https://www.aaii.org/ocs/index.php/ICWSM/ICWSM14/paper/viewPaper/8109>
- ▶ Daniel Jurafsky & James H. Martin (2021).
Chapter 20: Lexicons for Sentiment, Affect, and Connotation.
Speech and Language Processing, 3rd ed. draft.
<https://web.stanford.edu/~jurafsky/slp3/20.pdf>

Credit: slides based on Dr Loïc Barrault's slides for the autumn 2020/2021 course.

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In Proceedings of the 2004 ACM SIGKDD international conference on Knowledge discovery and data mining - KDD '04, page 168, New York, New York, USA. ACM Press.

Liu, B. (2011).

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Opinion observer: analyzing and comparing opinions on the Web.

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